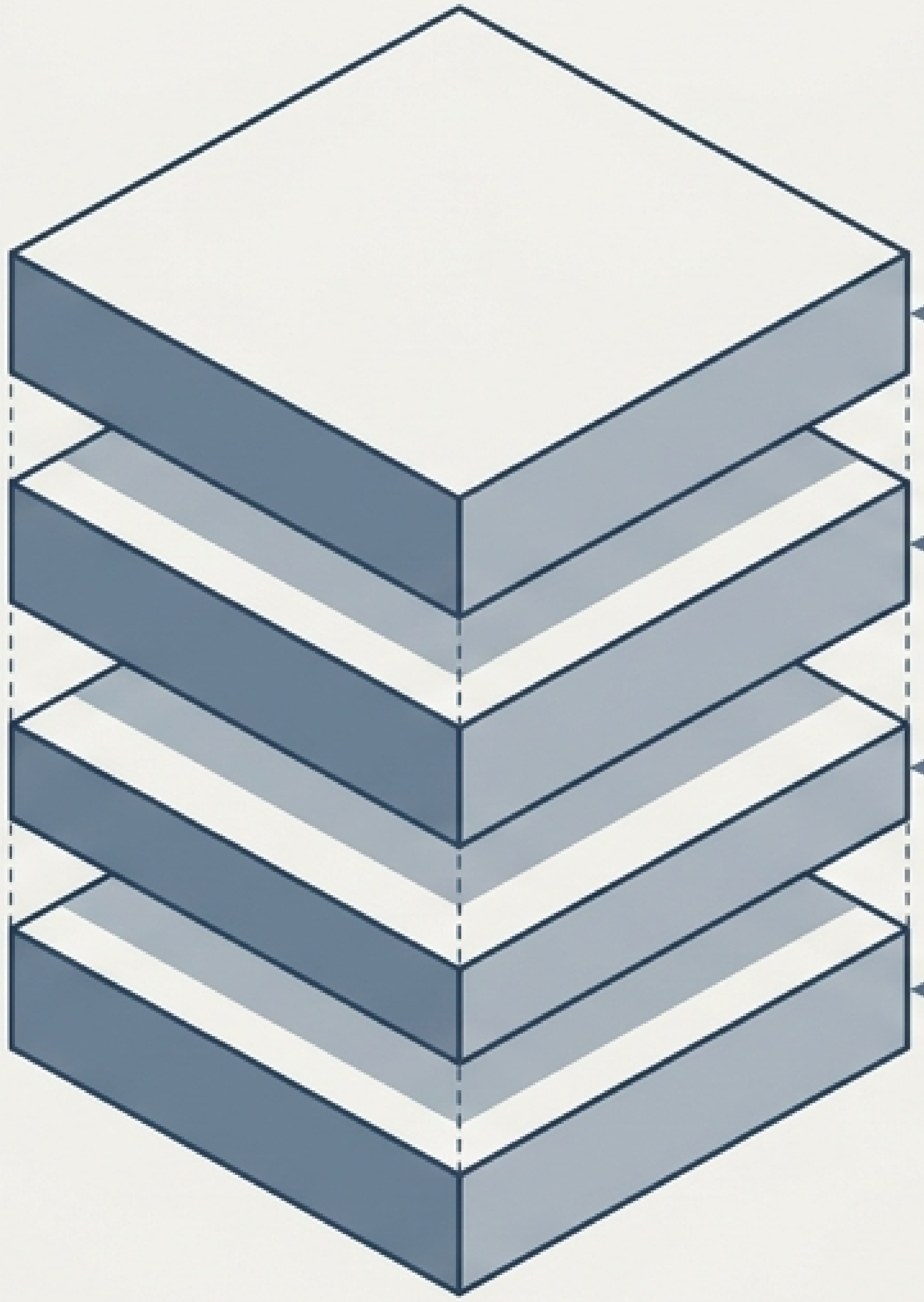
Hadoop Ecosystem Installation

Complete Data Engineer Blueprint: VMware, Ubuntu, Java 8, and Hadoop 2.7.3

Based on the Hadoop & Cloud Academy Masterclass.

Includes complete code-to-context translations and configuration matrices.

The Architecture Stack



Java 8 + Hadoop 2.7.3

Function: The distributed Big Data storage and processing framework.

Ubuntu Linux (64-bit)

Requirement: Minimum 20GB allocated disk space.
Function: The Unix-based guest OS strictly required by Hadoop.

VMware Workstation/Player

Function: The virtualization hypervisor bridging Windows and Linux.

Host Hardware

Requirement: Windows OS, Minimum 8GB RAM.
Function: The physical foundation.

Phase 1: Virtual Machine Provisioning

Hypervisor Provisioning

Install VMware Workstation (15/17) or Player on the host machine.

OS Injection

Select `Create New Virtual Machine` & browse to inject the downloaded Ubuntu 5.9GB ISO file.

Authentication Setup

Set username (lowercase only) and use ‘admin’ as the default system password.

Resource Allocation

Allocate minimum 20GB disk space. Leave all other hardware settings as default.

Note: Import process takes 45–60 minutes.

Phase 2: The Java 8 Dependency

```
~$ sudo apt update
~$ sudo apt install openjdk-8-jdk
~$ java -version
```

Fetch Repositories: Opens the Ubuntu terminal (**Ctrl + Alt + T**) and updates the package index. Requires the admin password.

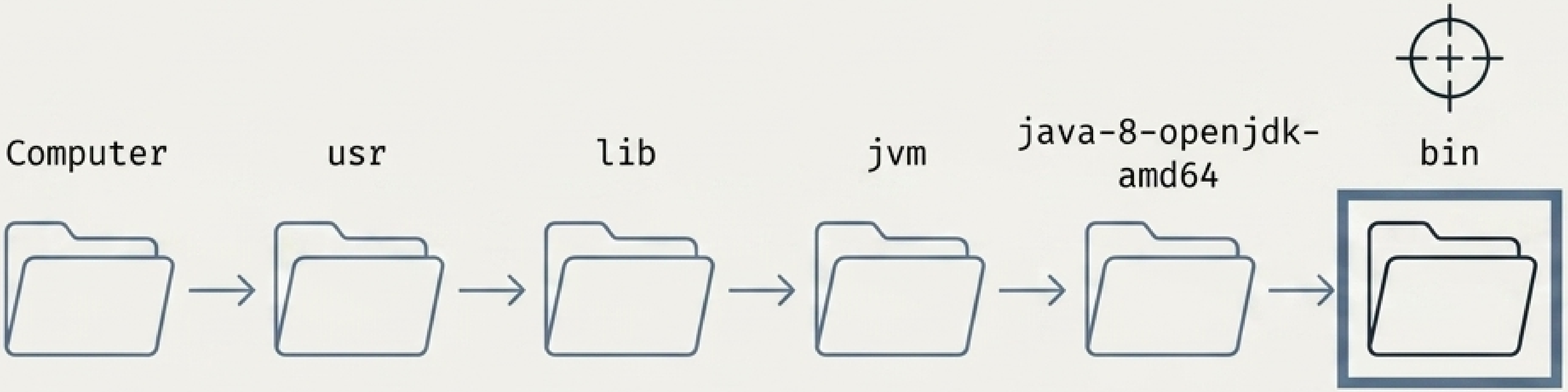
Install Java 8: Hadoop is written in Java and strictly requires version 8 (1.8.x). Type **'y'** when prompted to approve the download.

Verify: Confirms the successful installation of OpenJDK 1.8.x on the virtual machine.

Pro-Tip: Use **Ctrl + Shift + V** to paste copied text into the Ubuntu terminal.

Locating the Absolute Java Path

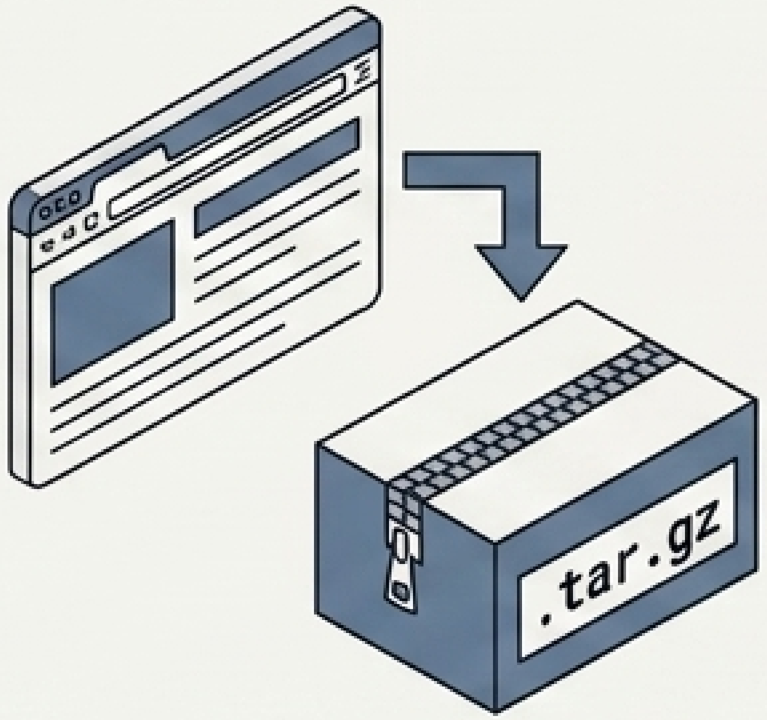
Hadoop requires the exact physical location of the Java Virtual Machine (JVM) to boot.



Action Required: Navigate to this directory using the Ubuntu file explorer. Right-click the "bin" folder, select Properties, and copy the exact path string. Save this path for the configuration phase.

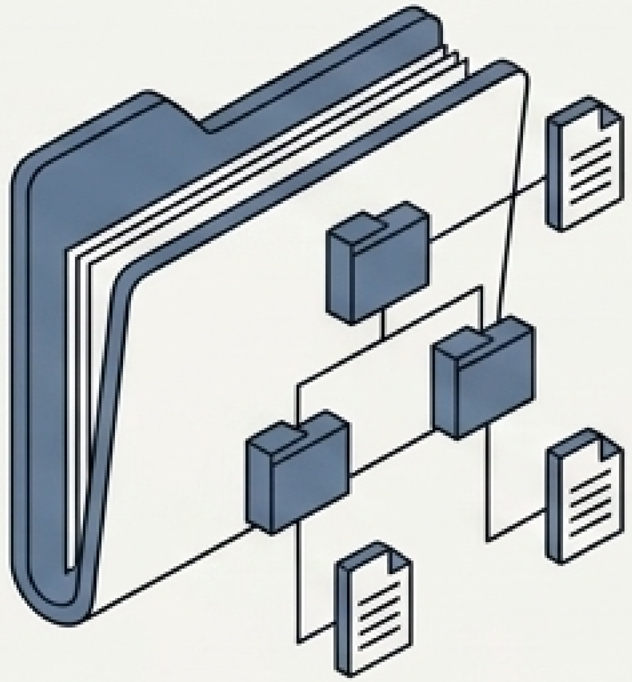
Phase 3: Downloading & Extracting Hadoop

The Payload



- **Source:** Open the Ubuntu browser and download **hadoop-2.7.3.tar.gz** (select the 204 MB version).
- **Location:** Move the **.tar.gz** file directly into your Ubuntu **Home directory**.

Extraction



- **Action:** Right-click the archive and select **Extract Here**.
- **Result:** Creates the active **hadoop-2.7.3 directory** inside **Home**. Do not rename the internal files.

Establishing Passwordless SSH

Hadoop nodes must communicate internally without prompting for passwords.

1. Local Connection

```
ssh localhost
```

Attempts a loopback connection. When prompted by the system security warning, type "yes" and press Enter.

2. Key Generation & Authorization

```
ssh-keygen
```

Generates an RSA keypair. Press Enter three consecutive times to leave passphrases empty.

```
cat ~/.ssh/id_rsa.pub >>
 ~/.ssh/authorized_keys
```

Appends the newly generated public key to the authorization list, bypassing future password prompts. Press **Ctrl + D** to exit.

Injecting Global Variables (.bashrc)

Run: **`sudo gedit ~/.bashrc`**

```
# User specific aliases and functions
# [End of original file]
export JAVA_HOME=/usr/lib/jvm/java-8-openjdk-amd64
export HADOOP_HOME=/home/username/hadoop-2.7.3
```

Replace ‘**username**’ with your exact Ubuntu profile name. Save, close the editor, and run ‘**source ~/.bashrc**’ in the terminal to immediately activate these variables.

The Core Configuration Matrix

Target Directory: `~/hadoop-2.7.3/etc/hadoop/`

File Name	Action Required
`hadoop-env.sh`	Replace the default `JAVA_HOME` variable on line 25 with the absolute `/usr/lib/jvm/...` path.
`core-site.xml`	Inject the HDFS default filesystem properties between the `<configuration>` tags.
`hdfs-site.xml`	Inject NameNode, DataNode, and replication properties between the `<configuration>` tags.
`mapred-site.xml`	First, duplicate and rename from `mapred-site.xml.template`. Then inject MapReduce framework properties.
`yarn-site.xml`	Inject YARN resource management properties.

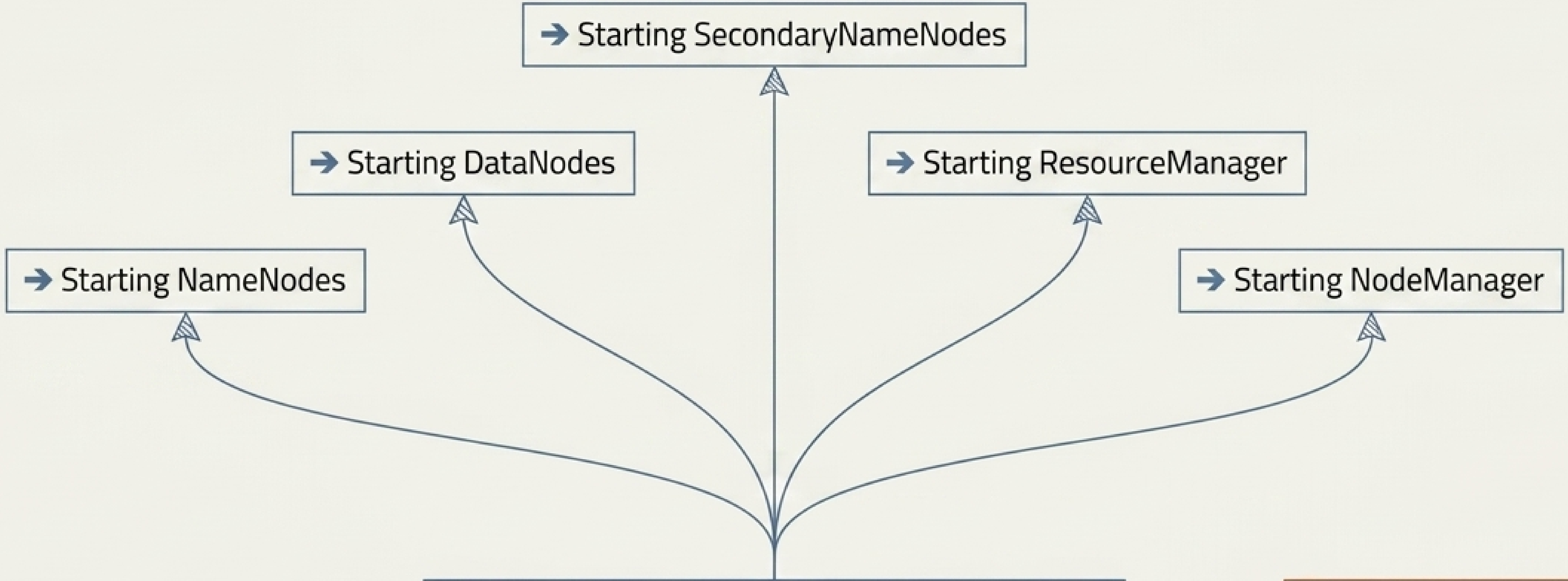
Phase 4: Formatting the NameNode

CRITICAL WARNING: This is a one-time initialization sequence. Never execute this command on a running system or you will wipe the entire Hadoop Distributed File System (HDFS).

```
1. cd /home/username/hadoop-2.7.3/bin
2. ./hadoop namenode -format
```

Success Indicator
Outcome: Look for the terminal output reading: “Storage directory has been successfully formatted.”

Booting the Cluster Services




```
cd /home/username/hadoop-2.7.3/sbin
./start-all.sh
```

Note:
If the system prompts for SSH authorization during this initial boot sequence, type 'yes'.

Verification: The JPS Diagnostic

Run the Java Virtual Machine Process Status Tool (`jps`) to confirm the Big Data Engine is active.

- [✔] NameNode
- [✔] DataNode
- [✔] SecondaryNameNode
- [✔] ResourceManager
- [✔] NodeManager

System Status: If exactly these five services appear (alongside the JPS process itself), the Hadoop architecture is fully installed, configured, and operational.

System Management Protocol

Performance Optimization

Host Cleanup: Uninstall unwanted background software on your host Windows machine to prevent hardware bottlenecks.

VM Tweaks: If Ubuntu lags heavily, power off the VM, navigate to VMware Edit Virtual Machine Settings, and increase RAM allocation to exactly 4GB.

Safe Shutdown Protocol

Rule: Never force-quit the VMware window using the operating system's 'X' button.

Action: Always use the top VMware menu. Click the Arrow Dropdown and select Shut Down Guest.

Alternative: Use Suspend only if pausing work temporarily.